

因子体系三层结构 - 简版

因子三层结构体系

ETHUSDT 1h · 训练集 55天 / 测试集 35天

结构说明

层级	名称	数量	说明
第一层	母因子主题 Parent Theme	~10	具有共同经济含义的因子家族（当前workspace有4个）
第二层	母因子 Parent Factor	4	可独立研究的核心逻辑单元
第三层	子因子 Sub Factor	257	具体公式实例

第一层：母因子主题

根据 docs/factor_structure.md 定义，共10个主题。当前workspace实现了其中4个：

主题ID	主题名称	状态	对应母因子
momentum	动量类 Momentum	✓ 已实现	momentum
volatility	波动率类 Volatility	✓ 已实现	volatility
flow	资金流类 Flow	✓ 已实现	flow
position_structure	持仓结构类 Position Structure	✓ 已实现	position_structure

第二层：母因子

momentum — 动量类 Momentum

假设：ETH 高频市场中，短期价格动量具有延续性。

- 1-4 小时窗口的动量效应较强
- 4-24 小时窗口存在均值回归

- 成交量放大的动量信号更可靠

子因子数量： 64 个 | 入选数量： 3 个

volatility — 波动率类 Volatility

假设： 波动率变化预示市场情绪转变。

- 短期波动率扩张往往伴随趋势加速
- 波动率压缩后突破具有方向性
- 高波动环境下反转策略有效

子因子数量： 64 个 | 入选数量： 3 个

flow — 资金流类 Flow

假设： 资金流向反映真实交易意图。

- 主动买入/卖出量失衡预示短期方向
- 大成交量伴随价格突破更可靠
- 资金费率变化反映多头/空头情绪

子因子数量： 65 个 | 入选数量： 3 个

position_structure — 持仓结构类 Position Structure

假设： 持仓结构变化反映市场参与者行为。

- 持仓量增加伴随价格上涨，多头强势
- 多空账户比极端值预示反转
- 持仓量快速减少往往伴随趋势结束

子因子数量： 64 个 | 入选数量： 6 个

第三层：子因子

momentum — 动量类 Momentum （64个子因子）

| 子因子ID | 名称 | 窗口 | 公式 | 入选 |

|-----|-----|-----|-----|-----|

| ret_1h | 1小时收益率 | 1h | close.pct_change(1) ||

| ret_4h | 4小时收益率 | 4h | close.pct_change(4) ||

| ret_8h | 8小时收益率 | 8h | close.pct_change(8) ||

ret_12h	12小时收益率	12h	close.pct_change(12)	
ret_24h	24小时收益率	24h	close.pct_change(24)	
ret_48h	48小时收益率	48h	close.pct_change(48)	
ret_72h	72小时收益率	72h	close.pct_change(72)	
log_ret_1h	1小时对数收益率	1h	np.log(close / close.shift(1))	
log_ret_4h	4小时对数收益率	4h	np.log(close / close.shift(4))	
log_ret_8h	8小时对数收益率	8h	np.log(close / close.shift(8))	
log_ret_24h	24小时对数收益率	24h	np.log(close / close.shift(24))	
log_ret_72h	72小时对数收益率	72h	np.log(close / close.shift(72))	
volume_momentum_4h	成交量动量(4h)	4h	volume.rolling(4).mean() / volume.rolling(12)...	
volume_momentum_8h	成交量动量(8h)	8h	volume.rolling(8).mean() / volume.rolling(24)...	
volume_momentum_12h	成交量动量(12h)	12h	volume.rolling(12).mean() / volume.rolling(36)...	
volume_momentum_24h	成交量动量(24h)	24h	volume.rolling(24).mean() / volume.rolling(72)...	
price_volume_corr_4h	量价相关性(4h)	4h	close.pct_change().rolling(4).corr(volume.pct...	✓
price_volume_corr_8h	量价相关性(8h)	8h	close.pct_change().rolling(8).corr(volume.pct...	✓
price_volume_corr_12h	量价相关性(12h)	12h	close.pct_change().rolling(12).corr(volume.pc...	
price_volume_corr_24h	量价相关性(24h)	24h	close.pct_change().rolling(24).corr(volume.pc...	✓
price_volume_corr_48h	量价相关性(48h)	48h	close.pct_change().rolling(48).corr(volume.pc...	
rsi_6	RSI(6)	6h	rsi(close, 6)	
rsi_14	RSI(14)	14h	rsi(close, 14)	
rsi_24	RSI(24)	24h	rsi(close, 24)	
taker_imbalance_4h	主动买卖失衡(4h)	4h	taker_buy_volume.rolling(4).sum() / volume.ro...	

```
| taker_imbalance_8h | 主动买卖失衡(8h) | 8h |  
taker_buy_volume.rolling(8).sum() / volume.ro... ||  
| taker_imbalance_12h | 主动买卖失衡(12h) | 12h |  
taker_buy_volume.rolling(12).sum() / volume.r... ||  
| taker_imbalance_24h | 主动买卖失衡(24h) | 24h |  
taker_buy_volume.rolling(24).sum() / volume.r... ||  
| taker_imbalance_48h | 主动买卖失衡(48h) | 48h |  
taker_buy_volume.rolling(48).sum() / volume.r... ||  
| momentum_reversal_4h | 动量反转(4h) | 4h | close.pct_change(4) -  
close.pct_change(4).rol... ||  
| close_ma_ratio_4h | 收盘均线比(4h) | 4h | close / close.rolling(4).mean() - 1  
||  
| close_ma_ratio_24h | 收盘均线比(24h) | 24h | close / close.rolling(24).mean()  
- 1 ||  
| close_ma_ratio_72h | 收盘均线比(72h) | 72h | close / close.rolling(72).mean()  
- 1 ||  
| vwap_deviation_8h | VWAP偏离(8h) | 8h | close / vwap(8) - 1 ||  
| vwap_deviation_24h | VWAP偏离(24h) | 24h | close / vwap(24) - 1 ||  
| cum_hourly_ret_24h | 累计小时收益(24h) | 24h |  
close.pct_change().rolling(24).sum() ||  
| cum_hourly_ret_72h | 累计小时收益(72h) | 72h |  
close.pct_change().rolling(72).sum() ||  
| ls_ratio_momentum_8h | 多空比动量(8h) | 8h | long_short_ratio.pct_change(8) ||  
| ls_ratio_momentum_24h | 多空比动量(24h) | 24h |  
long_short_ratio.pct_change(24) ||  
| ls_ratio_mean_reversion_24h | 多空比均值回复(24h) | 24h | long_short_ratio -  
long_short_ratio.rolling(2... ||  
| ls_ratio_zscore_24h | 多空比Z-score(24h) | 24h | (long_short_ratio -  
long_short_ratio.rolling(... ||  
| taker_bs_ratio_mom_8h | 主动买卖比动量(8h) | 8h |  
taker_buy_sell_ratio.pct_change(8) ||  
| taker_bs_ratio_zscore_24h | 主动买卖比Z-score(24h) | 24h |  
(taker_buy_sell_ratio - taker_buy_sell_ratio.... ||
```

```

| open_gap_momentum_1h | 开盘跳空动量(1h) | 1h | (open - close.shift(1)) /
close.shift(1) ||
| ret_acceleration_4h | 收益加速度(4h) | 4h | close.pct_change(1) -
close.pct_change(1).shi... ||
| risk_adj_momentum_24h | 风险调整动量(24h) | 24h | close.pct_change(24) /
(close.pct_change().ro... ||
| bb_width_zscore_8h | 布林Z-score(8h) | 8h | (close - close.rolling(8).mean())
/ close.rol... ||
| bb_width_zscore_24h | 布林Z-score(24h) | 24h | (close -
close.rolling(24).mean()) / close.ro... ||
| close_ts_rank_24h | 收盘时序分位(24h) | 24h | percentile_rank(close, 24) ||
| ret_ts_rank_24h | 收益时序分位(24h) | 24h |
percentile_rank(close.pct_change(), 24) ||
| rsi_slope_8h | RSI斜率(8h) | 8h | rsi(close, 14).diff(8) ||
| body_ratio_mean_8h | 实体占比均值(8h) | 8h | ((close - open).abs() /
close).rolling(8).mea... ||
| hl_range_momentum_8h | 振幅动量(8h) | 8h | ((high - low) /
close).rolling(8).mean().pct... ||
| taker_vol_ratio_momentum_8h | 主买主卖量比动量(8h) | 8h | (taker_buy_vol /
(taker_sell_vol + 1e-12)).ro... ||
| quote_volume_trend_8h | 成交额趋势比(8h) | 8h |
quote_volume.rolling(8).mean() / quote_volume... ||
| oi_value_price_divergence_24h | 名义持仓价格背离(24h) | 24h |
open_interest_value.pct_change(24) - close.pc... ||
| funding_signed_return_24h | 资金费方向对齐收益(24h) | 24h |
funding_rate.rolling(24).mean() * np.sign(clo... ||
| return_skew_momentum_24h | 收益偏度(24h) | 24h |
skewness(close.pct_change(), 24) ||
| return_kurt_momentum_24h | 收益峰度(24h) | 24h |
kurtosis(close.pct_change(), 24) ||
| long_account_momentum_8h | 多头账户动量(8h) | 8h | long_account.pct_change(8)
||
| short_account_mean_reversion_24h | 空头账户均值回复(24h) | 24h | short_account
- short_account.rolling(24).mea... ||

```

| account_spread_ma_24h | 多空账户差均值(24h) | 24h | (long_account - short_account).rolling(24).me... ||

| dist_below_high_24h | 距高点距离(24h) | 24h | (close - high.rolling(24).max().shift(1)) / c... ||

| dist_above_low_24h | 距低点距离(24h) | 24h | (close - low.rolling(24).min().shift(1)) / cl... ||

volatility — 波动率类 Volatility (64个子因子)

| 子因子ID | 名称 | 窗口 | 公式 | 入选 |

|-----|-----|-----|-----|-----|

| realized_vol_4h | 已实现波动率(4h) | 4h | close.pct_change().rolling(4).std() * np.sqrt... ||

| realized_vol_8h | 已实现波动率(8h) | 8h | close.pct_change().rolling(8).std() * np.sqrt... ||

| realized_vol_12h | 已实现波动率(12h) | 12h | close.pct_change().rolling(12).std() * np.sqr... ||

| realized_vol_24h | 已实现波动率(24h) | 24h | close.pct_change().rolling(24).std() * np.sqr... ||

| realized_vol_48h | 已实现波动率(48h) | 48h | close.pct_change().rolling(48).std() * np.sqr... ||

| realized_vol_72h | 已实现波动率(72h) | 72h | close.pct_change().rolling(72).std() * np.sqr... ||

| vol_ratio_1_24 | 波动率比(1h/24h) | 24h | close.pct_change().abs() / close.pct_change()... ||

| vol_ratio_4_24 | 波动率比(4h/24h) | 24h | close.pct_change().rolling(4).std() / close.p... ||

| vol_ratio_4_72 | 波动率比(4h/72h) | 72h | close.pct_change().rolling(4).std() / close.p... ||

| vol_ratio_8_72 | 波动率比(8h/72h) | 72h | close.pct_change().rolling(8).std() / close.p... ||

| vol_ratio_24_72 | 波动率比(24h/72h) | 72h | close.pct_change().rolling(24).std() / close.... ||

| high_low_range_4h | 高低价幅度(4h) | 4h | (high.rolling(4).max() - low.rolling(4).min())... ||

| high_low_range_8h | 高低价幅度(8h) | 8h | (high.rolling(8).max() - low.rolling(8).min())... ||

| high_low_range_12h | 高低价幅度(12h) | 12h | (high.rolling(12).max() - low.rolling(12).min())... ||

| high_low_range_24h | 高低价幅度(24h) | 24h | (high.rolling(24).max() - low.rolling(24).min())... ||

| high_low_range_48h | 高低价幅度(48h) | 48h | (high.rolling(48).max() - low.rolling(48).min())... ||

| close_position_4h | 收盘位置(4h) | 4h | (close - low.rolling(4).min()) / (high.rollin... ||

| close_position_8h | 收盘位置(8h) | 8h | (close - low.rolling(8).min()) / (high.rollin... | ✓ |

| close_position_12h | 收盘位置(12h) | 12h | (close - low.rolling(12).min()) / (high.rolli... ||

| close_position_24h | 收盘位置(24h) | 24h | (close - low.rolling(24).min()) / (high.rolli... ||

| quote_volume_zscore_8h | 成交额Z-score(8h) | 8h | (quote_volume - quote_volume.rolling(8).mean(... ||

| quote_volume_zscore_24h | 成交额Z-score(24h) | 24h | (quote_volume - quote_volume.rolling(24).mean(... ||

| quote_volume_zscore_72h | 成交额Z-score(72h) | 72h | (quote_volume - quote_volume.rolling(72).mean(... ||

| parkinson_vol_8h | Parkinson波动率(8h) | 8h | np.sqrt((np.log(high/low)**2).rolling(8).mean(... ||

| parkinson_vol_24h | Parkinson波动率(24h) | 24h | np.sqrt((np.log(high/low)**2).rolling(24).mea... ||

| parkinson_vol_72h | Parkinson波动率(72h) | 72h | np.sqrt((np.log(high/low)**2).rolling(72).mea... ||

| vol_change_4h | 波动率变化(4h) | 4h | close.pct_change().rolling(4).std().pct_chang... | ✓ |

| vol_change_8h | 波动率变化(8h) | 8h | close.pct_change().rolling(8).std().pct_chang... ||

| vol_change_24h | 波动率变化(24h) | 24h | close.pct_change().rolling(24).std().pct_chan... | ✓ |

```

| intraday_range_1h | 日内振幅(1h) | 1h | (high - low) /
close.rolling(1).mean() ||

| gk_volatility_8h | GK波动率(8h) | 8h | np.sqrt(np.maximum(0, (0.5 *
(np.log(high / l... ||

| gk_volatility_24h | GK波动率(24h) | 24h | np.sqrt(np.maximum(0, (0.5 *
(np.log(high / l... ||

| gk_volatility_72h | GK波动率(72h) | 72h | np.sqrt(np.maximum(0, (0.5 *
(np.log(high / l... ||

| rs_volatility_8h | RS波动率(8h) | 8h | np.sqrt((np.log(high / close) *
np.log(high / ... ||

| rs_volatility_24h | RS波动率(24h) | 24h | np.sqrt((np.log(high / close) *
np.log(high / ... ||

| atr_ratio_8h | ATR占比(8h) | 8h | truerange.rolling(8).mean() / close ||

| atr_ratio_24h | ATR占比(24h) | 24h | truerange.rolling(24).mean() / close ||

| vol_term_structure_24h | 波动率期限斜率(24h) | 24h |
close.pct_change().rolling(8).std() - close.p... ||

| funding_rate_vol_24h | 资金费率波动(24h) | 24h |
funding_rate.rolling(24).std() ||

| ls_ratio_realized_vol_24h | 多空比实现波动(24h) | 24h |
long_short_ratio.pct_change().rolling(24).std... ||

| volume_log_vol_24h | 成交量对数波动(24h) | 24h | np.log(volume /
volume.shift(1).replace(0, np... ||

| oi_change_vol_24h | 持仓变化波动(24h) | 24h |
open_interest.pct_change().rolling(24).std() ||

| downside_vol_24h | 下行波动(24h) | 24h |
close.pct_change().clip(upper=0).rolling(24).... ||

| upside_vol_24h | 上行波动(24h) | 24h |
close.pct_change().clip(lower=0).rolling(24).... ||

| vol_asymmetry_24h | 波动不对称(24h) | 24h |
close.pct_change().clip(lower=0).rolling(24).... ||

| range_over_atr_24h | 区间ATR比(24h) | 24h | (high.rolling(24).max() -
low.rolling(24).min... ||

| down_hour_fraction_24h | 下跌时长占比(24h) | 24h | count(close.pct_change() <
0, 24) / 24 ||

```



```

| vwap_dispersion_vol_24h | VWAP偏离波动(24h) | 24h | (close -
vwap(24)).rolling(24).std() / close ||

| quote_volume_cv_24h | 成交额变异系数(24h) | 24h |
quote_volume.rolling(24).std() / (quote_volum... ||

| taker_imbalance_vol_24h | 主动买入占比波动(24h) | 24h | (taker_buy_volume /
(volume + 1e-12)).rolling... ||

| body_ratio_vol_8h | 实体占比波动(8h) | 8h | ((close - open).abs() /
close).rolling(8).std... ||

| upper_shadow_vol_8h | 上影线波动(8h) | 8h | ((high - np.maximum(open,
close)) / close).ro... ||

| quote_price_vol_ratio_24h | 额价波动比(24h) | 24h |
quote_volume.pct_change().rolling(24).std() / ... ||

| oi_price_vol_product_24h | 持仓价格波动乘积(24h) | 24h |
open_interest.pct_change().abs().rolling(24).... ||

| funding_abs_mean_24h | 资金费绝对均值(24h) | 24h |
funding_rate.abs().rolling(24).mean() ||

| log_return_std_8h | 对数收益标准差(8h) | 8h | np.log(close /
close.shift(1)).rolling(8).std... ||

| vol_cluster_ratio_8h | 波动聚集比(8h) | 8h |
close.pct_change().rolling(8).std() / (close.... ||

| oi_value_change_vol_24h | 名义持仓波动(24h) | 24h |
open_interest_value.pct_change().rolling(24).... ||

| taker_bs_ratio_realized_vol_24h | 主买主卖比实现波动(24h) | 24h |
taker_buy_sell_ratio.pct_change().rolling(24)... ||

| long_account_vol_24h | 多头账户波动(24h) | 24h |
long_account.pct_change().rolling(24).std() ||

| short_account_vol_24h | 空头账户波动(24h) | 24h |
short_account.pct_change().rolling(24).std() ||

| return_skew_vol_24h | 收益偏度波动因子(24h) | 24h |
skewness(close.pct_change(), 24) ||

| return_kurt_vol_24h | 收益峰度波动因子(24h) | 24h |
kurtosis(close.pct_change(), 24) ||

| ret_range_corr_24h | 收益振幅相关(24h) | 24h |
close.pct_change().rolling(24).corr((high - l... ||

```

flow — 资金流类 Flow (65个子因子)

| 子因子ID | 名称 | 窗口 | 公式 | 入选 |

|-----|-----|-----|-----|-----|

| funding_rate_raw | 资金费率 (原始) | 1h | funding_rate | ☒ |

| funding_rate_ma_8h | 资金费率均值(8h) | 8h | funding_rate.rolling(8).mean() ||

| funding_rate_ma_24h | 资金费率均值(24h) | 24h |

funding_rate.rolling(24).mean() ||

| funding_rate_ma_48h | 资金费率均值(48h) | 48h |

funding_rate.rolling(48).mean() ||

| funding_rate_ma_72h | 资金费率均值(72h) | 72h |

funding_rate.rolling(72).mean() ||

| funding_rate_zscore_24h | 资金费率Z-score(24h) | 24h | (funding_rate -

funding_rate.rolling(24).mean... | ☒ |

| funding_rate_zscore_72h | 资金费率Z-score(72h) | 72h | (funding_rate -

funding_rate.rolling(72).mean... ||

| funding_rate_change_8h | 资金费率变化(8h) | 8h | funding_rate.pct_change(8) ||

| funding_rate_change_24h | 资金费率变化(24h) | 24h |

funding_rate.pct_change(24) ||

| funding_rate_change_72h | 资金费率变化(72h) | 72h |

funding_rate.pct_change(72) ||

| funding_rate_acceleration_8h | 资金费率加速度(8h) | 8h |

funding_rate.diff(8).diff(8) | ☒ |

| funding_rate_acceleration_24h | 资金费率加速度(24h) | 24h |

funding_rate.diff(24).diff(24) ||

| taker_buy_ratio_1h | 主动买入比(1h) | 1h | taker_buy_volume.rolling(1).sum()

/ volume.ro... ||

| taker_buy_ratio_4h | 主动买入比(4h) | 4h | taker_buy_volume.rolling(4).sum()

/ volume.ro... ||

| taker_buy_ratio_8h | 主动买入比(8h) | 8h | taker_buy_volume.rolling(8).sum()

/ volume.ro... ||

| taker_buy_ratio_24h | 主动买入比(24h) | 24h |

taker_buy_volume.rolling(24).sum() / volume.r... ||

| taker_buy_ratio_72h | 主动买入比(72h) | 72h |

taker_buy_volume.rolling(72).sum() / volume.r... ||

```

| taker_buy_amount_ratio_1h | 主动买入额占比(1h) | 1h |
taker_buy_quote_volume.rolling(1).sum() / quo... ||

| taker_buy_amount_ratio_4h | 主动买入额占比(4h) | 4h |
taker_buy_quote_volume.rolling(4).sum() / quo... ||

| taker_buy_amount_ratio_8h | 主动买入额占比(8h) | 8h |
taker_buy_quote_volume.rolling(8).sum() / quo... ||

| taker_buy_amount_ratio_24h | 主动买入额占比(24h) | 24h |
taker_buy_quote_volume.rolling(24).sum() / qu... ||

| taker_buy_amount_ratio_72h | 主动买入额占比(72h) | 72h |
taker_buy_quote_volume.rolling(72).sum() / qu... ||

| volume_change_1h | 成交量变化(1h) | 1h | volume.pct_change(1) ||
| volume_change_4h | 成交量变化(4h) | 4h | volume.pct_change(4) ||
| volume_change_8h | 成交量变化(8h) | 8h | volume.pct_change(8) ||
| volume_change_24h | 成交量变化(24h) | 24h | volume.pct_change(24) ||
| volume_change_72h | 成交量变化(72h) | 72h | volume.pct_change(72) ||

| volume_zscore_8h | 成交量Z-score(8h) | 8h | (volume -
volume.rolling(8).mean()) / volume.... ||

| volume_zscore_24h | 成交量Z-score(24h) | 24h | (volume -
volume.rolling(24).mean()) / volume... ||

| volume_zscore_72h | 成交量Z-score(72h) | 72h | (volume -
volume.rolling(72).mean()) / volume... ||

| funding_oi_value_pressure_8h | 资金费持仓名义压力(8h) | 8h | (funding_rate *
open_interest_value).rolling(... ||

| funding_oi_value_pressure_24h | 资金费持仓名义压力(24h) | 24h | (funding_rate
* open_interest_value).rolling(... ||

| volume_oi_turnover_1h | 成交量持仓比(1h) | 1h | volume / (open_interest + 1e-
12) ||

| volume_oi_turnover_8h | 成交量持仓比(8h) | 8h | volume.rolling(8).mean() /
(open_interest.rol... ||

| volume_oi_turnover_24h | 成交量持仓比(24h) | 24h | volume.rolling(24).sum() /
(open_interest.rol... ||

| quote_oi_value_ratio_1h | 成交额名义持仓比(1h) | 1h | quote_volume /
(open_interest_value + 1e-12) ||

```

| quote_oi_value_ratio_8h | 成交额名义持仓比(8h) | 8h |
quote_volume.rolling(8).sum() / (open_interes... ||

| taker_price_size_spread_8h | 主动买卖价差(8h) | 8h |
(taker_buy_quote_volume / (quote_volume + 1e-... ||

| funding_volume_z_interaction_24h | 资金费成交量Z交互(24h) | 24h |
((funding_rate - funding_rate.rolling(24).mea... ||

| ls_funding_flow_24h | 多空比资金费流(24h) | 24h | (long_short_ratio *
funding_rate).rolling(24)... ||

| vwap_volume_pressure_8h | VWAP成交量压力(8h) | 8h | ((close - vwap(8)) *
volume).rolling(8).sum()... ||

| taker_vol_ratio_mr_24h | 主买主卖量比均值回复(24h) | 24h | (taker_buy_vol /
(taker_sell_vol + 1e-12)) - ... ||

| funding_return_alignment_8h | 资金费收益协同(8h) | 8h |
funding_rate.pct_change(8) * close.pct_change... ||

| volume_oi_flow_divergence_8h | 成交量持仓流背离(8h) | 8h |
volume.pct_change(8) - open_interest.pct_chan... ||

| quote_oi_value_divergence_24h | 成交额名义持仓背离(24h) | 24h |
quote_volume.pct_change(24) - open_interest_v... ||

| taker_amount_volume_corr_24h | 主动额量占比相关(24h) | 24h |
(taker_buy_quote_volume / (quote_volume + 1e-... ||

| funding_log_return_corr_24h | 资金费对数收益相关(24h) | 24h |
funding_rate.rolling(24).corr(np.log(close / ... ||

| account_spread_volume_24h | 账户差成交量(24h) | 24h | ((long_account -
short_account) * volume).rol... ||

| taker_bs_ratio_level_24h | 主动买卖比水平(24h) | 24h |
taker_buy_sell_ratio.rolling(24).mean() ||

| taker_bs_ratio_trend_8h | 主动买卖比趋势(8h) | 8h |
taker_buy_sell_ratio.rolling(8).mean() / (tak... ||

| oi_contract_value_drift_24h | 张数名义比漂移(24h) | 24h | (open_interest /
(open_interest_value + 1e-12... ||

| funding_ls_cov_proxy_24h | 资金费多空协方差代理(24h) | 24h | (funding_rate *
long_short_ratio).rolling(24)... ||

| volume_abs_ret_cohesion_24h | 成交量绝对收益凝聚(24h) | 24h | (volume *
close.pct_change().abs()).rolling(2... ||

taker_sum_ratio_momentum_24h	主动买入累计占比动量(24h) 24h	$(\text{taker_buy_volume.rolling}(24).sum() / (\text{volume} \dots))$	
funding_volume_sign_product_1h	资金费成交量符号积(1h) 1h	$\text{np.sign}(funding_rate) * \text{np.sign}(\text{volume.pct_ch} \dots)$	
passive_quote_fraction_1h	非主动成交额占比(1h) 1h	$(\text{quote_volume} - \text{taker_buy_quote_volume}) / (\text{qu} \dots)$	
passive_quote_fraction_24h	非主动成交额占比(24h) 24h	$((\text{quote_volume} - \text{taker_buy_quote_volume}) / (\text{q} \dots))$	
ls_volume_interaction_8h	多空比成交量交互(8h) 8h	$\text{long_short_ratio} * \text{volume.pct_change}(8)$	
funding_per_oi_value_1h	单位名义持仓资金费(1h) 1h	$\text{funding_rate} / (\text{open_interest_value} + 1e-12)$	
ret_volume_change_corr_24h	收益成交量变化相关(24h) 24h	$\text{close.pct_change().rolling}(24).corr(\text{volume.pc} \dots)$	
taker_share_trend_24h	主动买入份额趋势(24h) 24h	$(\text{taker_buy_volume.rolling}(24).mean() / (\text{volum} \dots))$	
funding_oi_change_product_24h	资金费持仓变化乘积(24h) 24h	$(\text{funding_rate} * \text{open_interest.pct_change()}).$	
vwap_taker_combo_8h	VWAP主动买入组合(8h) 8h	$((\text{close} / (\text{vwap}(8) + 1e-12) - 1) * (\text{taker_buy} \dots))$	
long_account_funding_spread_24h	多头占比资金费差(24h) 24h	$\text{long_account} - \text{funding_rate.rolling}(24).mean(\dots)$	
short_account_funding_combo_24h	空头占比资金费组合(24h) 24h	$\text{short_account} + \text{funding_rate.rolling}(24).mean(\dots)$	

position_structure — 持仓结构类 Position Structure（64个子因子）

子因子ID	名称	窗口	公式	入选
oi_raw	持仓量（原始） 1h	open_interest	☒	
oi_change_1h	持仓量变化(1h) 1h	open_interest.pct_change(1)		
oi_change_4h	持仓量变化(4h) 4h	open_interest.pct_change(4)		
oi_change_8h	持仓量变化(8h) 8h	open_interest.pct_change(8)	☒	
oi_change_24h	持仓量变化(24h) 24h	open_interest.pct_change(24)		
oi_change_72h	持仓量变化(72h) 72h	open_interest.pct_change(72)	☒	

| oi_ma_8h | 持仓量均值(8h) | 8h | open_interest.rolling(8).mean() ||

| oi_ma_24h | 持仓量均值(24h) | 24h | open_interest.rolling(24).mean() ||

| oi_ma_72h | 持仓量均值(72h) | 72h | open_interest.rolling(72).mean() ||

| oi_zscore_24h | 持仓量Z-score(24h) | 24h | (open_interest - open_interest.rolling(24).me... ||

| oi_zscore_72h | 持仓量Z-score(72h) | 72h | (open_interest - open_interest.rolling(72).me... | ☒ |

| oi_price_divergence_4h | 持仓量价格背离(4h) | 4h | open_interest.pct_change(4) - close.pct_chang... ||

| oi_price_divergence_8h | 持仓量价格背离(8h) | 8h | open_interest.pct_change(8) - close.pct_chang... ||

| oi_price_divergence_24h | 持仓量价格背离(24h) | 24h | open_interest.pct_change(24) - close.pct_chan... ||

| oi_price_divergence_72h | 持仓量价格背离(72h) | 72h | open_interest.pct_change(72) - close.pct_chan... ||

| oi_price_corr_8h | 持仓量价格相关性(8h) | 8h | open_interest.pct_change().rolling(8).corr(cl... ||

| oi_price_corr_24h | 持仓量价格相关性(24h) | 24h | open_interest.pct_change().rolling(24).corr(c... | ☒ |

| oi_price_corr_72h | 持仓量价格相关性(72h) | 72h | open_interest.pct_change().rolling(72).corr(c... ||

| oi_volume_ratio_8h | 持仓量成交比(8h) | 8h | open_interest.rolling(8).mean() / volume.roll... ||

| oi_volume_ratio_24h | 持仓量成交比(24h) | 24h | open_interest.rolling(24).mean() / volume.rol... | ☒ |

| oi_volume_ratio_72h | 持仓量成交比(72h) | 72h | open_interest.rolling(72).mean() / volume.rol... ||

| oi_acceleration_4h | 持仓量加速度(4h) | 4h | open_interest.diff(4).diff(4) ||

| oi_acceleration_8h | 持仓量加速度(8h) | 8h | open_interest.diff(8).diff(8) ||

| oi_acceleration_24h | 持仓量加速度(24h) | 24h | open_interest.diff(24).diff(24) ||

| oi_usd_change_1h | 持仓量USD变化(1h) | 1h | open_interest_value.pct_change(1) ||

| oi_usd_change_4h | 持仓量USD变化(4h) | 4h |
open_interest_value.pct_change(4) ||

| oi_usd_change_8h | 持仓量USD变化(8h) | 8h |
open_interest_value.pct_change(8) ||

| oi_usd_change_24h | 持仓量USD变化(24h) | 24h |
open_interest_value.pct_change(24) ||

| oi_usd_change_72h | 持仓量USD变化(72h) | 72h |
open_interest_value.pct_change(72) ||

| oi_buildup_8h | 持仓量累积(8h) | 8h | count(open_interest >
open_interest.shift(1),... ||

| oi_contract_per_value_1h | 张数名义持仓比(1h) | 1h | open_interest /
(open_interest_value + 1e-12) ||

| oi_contract_per_value_mr_24h | 张数名义比均值回复(24h) | 24h | (open_interest
/ (open_interest_value + 1e-12... ||

| oi_value_volume_ratio_1h | 名义持仓成交量比(1h) | 1h | open_interest_value /
(volume + 1e-12) ||

| oi_value_volume_ratio_8h | 名义持仓成交量比(8h) | 8h |
open_interest_value.rolling(8).mean() / (volu... ||

| ls_oi_product_24h | 多空比持仓乘积(24h) | 24h | (long_short_ratio *
open_interest).rolling(24... ||

| long_account_oi_corr_24h | 多头占比持仓变化相关(24h) | 24h |
long_account.rolling(24).corr(open_interest.p... ||

| short_account_oi_corr_24h | 空头占比持仓变化相关(24h) | 24h |
short_account.rolling(24).corr(open_interest.... ||

| funding_oi_change_corr_24h | 资金费持仓变化相关(24h) | 24h |
funding_rate.rolling(24).corr(open_interest.p... ||

| oi_value_funding_product_24h | 名义持仓变化资金费乘积(24h) | 24h |
(open_interest_value.pct_change() * funding_r... ||

| oi_price_zscore_spread_24h | 持仓价格Z差(24h) | 24h | ((open_interest -
open_interest.rolling(24).m... ||

| account_spread_per_oi_1h | 账户差每单位持仓(1h) | 1h | (long_account -
short_account) / (open_intere... ||

| taker_bs_oi_level_corr_24h | 主买主卖比持仓相关(24h) | 24h |
taker_buy_sell_ratio.rolling(24).corr(open_in... ||

| oi_ret_log_spread_24h | 持仓变化对数收益差(24h) | 24h |
open_interest.pct_change(24) - np.log(close / ... ||

| oi_value_price_ratio_1h | 名义持仓价格比(1h) | 1h | open_interest_value /
(close + 1e-12) ||

| oi_value_price_ratio_z_24h | 名义持仓价格比Z(24h) | 24h |
((open_interest_value / (close + 1e-12)) - (o... ||

| oi_taker_buy_sum_8h | 持仓主买量和(8h) | 8h | (open_interest +
taker_buy_volume).rolling(8)... ||

| oi_taker_share_change_corr_24h | 持仓主买占比变化相关(24h) | 24h |
open_interest.pct_change().rolling(24).corr(... ||

| ls_oi_change_spread_8h | 多空比持仓变化差(8h) | 8h |
long_short_ratio.pct_change(8) - open_interes... ||

| oi_value_per_ls_ratio_1h | 名义持仓每单位多空比(1h) | 1h | open_interest_value
/ (long_short_ratio + 1e-... ||

| oi_quote_turnover_ratio_1h | 持仓成交额比(1h) | 1h | open_interest /
(quote_volume + 1e-12) ||

| oi_quote_turnover_ratio_24h | 持仓成交额比(24h) | 24h |
open_interest.rolling(24).mean() / (quote_vol... ||

| oi_vwap_mispricing_24h | 持仓VWAP错价(24h) | 24h | (open_interest * (close /
(vwap(24) + 1e-12) ... ||

| long_short_account_ratio_1h | 多空账户比(1h) | 1h | long_account /
(short_account + 1e-12) ||

| long_short_account_ratio_mom_24h | 多空账户比动量(24h) | 24h | (long_account
/ (short_account + 1e-12)).pct_... ||

| oi_range_interaction_24h | 持仓振幅交互(24h) | 24h | (open_interest * ((high
- low) / (close + 1e-... ||

| oi_value_taker_bs_corr_24h | 名义持仓主买主卖比相关(24h) | 24h |
open_interest_value.pct_change().rolling(24).... ||

| oi_funding_abs_exposure_24h | 持仓绝对资金费暴露(24h) | 24h | (open_interest *
funding_rate.abs()).rolling(... ||

| oi_vs_oi_value_change_spread_24h | 张数名义持仓变化差(24h) | 24h |
open_interest.pct_change(24) - open_interest_... ||

| oi_volume_level_corr_24h | 持仓成交量水平相关(24h) | 24h |
open_interest.rolling(24).corr(volume) ||

| `account_sum_deviation_1h` | 多空账户和偏离(1h) | 1h | `long_account + short_account - 1` ||

| `oi_price_cov_proxy_24h` | 持仓价格协方差代理(24h) | 24h | `(open_interest * close).rolling(24).mean() - ...` ||

| `taker_buy_per_oi_mom_8h` | 主买每单位持仓动量(8h) | 8h | `(taker_buy_volume / (open_interest + 1e-12))....` ||

| `oi_value_per_taker_sell_1h` | 名义持仓每单位主卖量(1h) | 1h | `open_interest_value / (taker_sell_vol + 1e-12...` ||

| `oi_ls_zscore_pressure_8h` | 持仓多空Z压力(8h) | 8h | `(open_interest * (long_short_ratio - long_sho...` ||

入选因子汇总（15个）

| 子因子 | 所属母因子 | 综合评分 | IC方向 |

|-----|-----|-----|-----|

| `funding_rate_zscore_24h` | flow | 0.756 | + |

| `price_volume_corr_8h` | momentum | 0.646 | + |

| `funding_rate_raw` | flow | 0.642 | + |

| `funding_rate_acceleration_8h` | flow | 0.633 | + |

| `price_volume_corr_24h` | momentum | 0.578 | + |

| `oi_change_72h` | position_structure | 0.576 | - |

| `oi_raw` | position_structure | 0.561 | - |

| `oi_zscore_72h` | position_structure | 0.528 | - |

| `oi_price_corr_24h` | position_structure | 0.487 | + |

| `price_volume_corr_4h` | momentum | 0.473 | + |

| `close_position_8h` | volatility | 0.447 | - |

| `vol_change_24h` | volatility | 0.429 | + |

| `oi_volume_ratio_24h` | position_structure | 0.414 | + |

| `vol_change_4h` | volatility | 0.411 | + |

| `oi_change_8h` | position_structure | 0.403 | - |